

Literature-Implied Status for the Classical-Space Subquestion in arXiv:math/0304018

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July 3, 2026

Status. This note records a literature-implied answer to the classical-space subquestion in Ferenczi–Rosendal’s *Ergodic Banach Spaces*, arXiv:math/0304018. It is not a new result of the run. The broad conjecture that ℓ_2 is the only non-ergodic separable Banach space remains open in the later literature cited here.

Source question. Near the end of arXiv:math/0304018, after Proposition 20 and before the acknowledgements, Ferenczi and Rosendal write that they conjecture that ℓ_2 is the only non-ergodic Banach space. They then point out a concrete missing case: they are not even able to prove that c_0 and ℓ_p , $p \neq 2$, are ergodic.

Supporting theorem and identification. Cuellar Carrera’s *Non-ergodic Banach spaces are near Hilbert*, arXiv:1611.05500, proves in Theorem 3.2 of that paper that every separable non-near-Hilbert Banach space satisfies the Cantorized-Enflo criterion and is therefore ergodic. Equivalently, every non-ergodic separable Banach space must be near Hilbert. Since ℓ_p is not near Hilbert for $p \neq 2$, this theorem already implies the ℓ_p cases conceptually.

The paper also records the classical cases explicitly. In its introduction it cites Ferenczi–Galego for ergodicity of c_0 and ℓ_p for $1 \leq p < 2$. Its Corollary after Proposition 2.2 says that if ℓ_p , $p > 2$, is crudely finitely representable in a separable Banach space X , then X is ergodic; applying this to $X = \ell_p$ gives ergodicity for every $2 < p < \infty$. The same introduction explicitly says that this solves the Ferenczi–Rosendal question about the ergodicity of the ℓ_p spaces for $p > 2$.

Scope. Thus the explicit source subquestion about the classical spaces c_0 and ℓ_p , $p \neq 2$, is answered by known literature. This packet should prevent future agents from treating that subquestion as open. It does not settle the full conjecture that ℓ_2 is the only non-ergodic Banach space: Cuellar Carrera’s final remarks reduce the remaining territory to near-Hilbert phenomena and list HAP and twisted-Hilbert questions as natural remaining problems.

References.

- V. Ferenczi and C. Rosendal, *Ergodic Banach Spaces*, arXiv:math/0304018; Adv. Math. 195 (2005), 259–282.
- W. Cuellar Carrera, *Non-ergodic Banach spaces are near Hilbert*, arXiv:1611.05500.
- V. Ferenczi and E. Galego, *Some equivalence relations which are Borel reducible to isomorphism between separable Banach spaces*, Israel J. Math. 152 (2006), 61–82.